

WP-MIP GFDL Vortex Tracker Outputs (Version 0.0)

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This dataset represents the initial release (Version 0.0) of GFDL Vortex Tracker (Marchok 2021) outputs, which were made available via GitHub in May 2026 specifically for the WP-MIP. Comprehensive details regarding the dataset and its output formats are provided in the accompanying documentation, "GFDL Vortex Tracker Format Instructions for WP-MIP".

The model indices and contributing organizations are summarized in Table 1. As of mid-May 2026, the Vortex Tracker has been processed for all accessible WP-MIP model forecasts, encompassing both the Core and SP1 datasets. The initialization frequency for each model—either daily or every three days—is represented by the color coding within the table cells.

Identifier	Contributor	Forecasts OIC			Forecasts SIC		
		AI	Hybrid	Physical	AI	Hybrid	Physical
babj	CMA	cmao					
cwao	ECCC	cwao	cwho	cwpo	cwas	cwhs	cwps
ecmf	ECMWF	ecao	echo	ecpo			
edzw	DWD			dwpo			
egrr	UKMO	ukao		ukpo			ukps
gfdl	GFDL			shpo			shps
kwbc	NOAA	gfao		gfpo	gfas		gfps
rjtd	JMA	jmao		jmpo			
rksl	KMA/KIAPS			kmpo			
rums	RAS						
sbsj	CPTEC/INPE						

Table 1: Key to model indices for WP-MIP GFDL Vortex Tracker outputs. Cell coloring identifies initialization frequency: light green for daily, dark green for three-day intervals. Red indicates tracker runs are inhibited by missing essential variables; yellow denotes that data is provided and pending processing. Grey cells denote instances where contributors did not provide forecast data.

Tropical cyclones are tracked using metrics based on the following variables provided by each model: sea level pressure, u- and v-component winds at 10 m, 850 and 700 hPa, and geopotential heights at 850 and 700 hPa. The tracker also provides complete thermodynamic

diagnostics of Cyclone Phase Space (Hart 2003) and a check for a warm core in the 300-500 mb layer. These diagnostics are based on the geopotential heights at 925, 850, 700, 500, 400, and 300 hPa, as well as the temperature fields at 500, 400, and 300 hPa. The columns for these diagnostics will be left blank if a model does not provide all of the above mentioned variables.

Please check additional notes for some of the models below.

echo: Model version of “hy_00” used in this release.

ecpo: Tracker outputs will be updated to use daily forecasts in the next release.

jmao, jmpo, gfpo: Thermodynamic diagnostics at all or some lead times are not provided due to the lack of required variables or inadequate forecast interval.

References:

Hart, R. E., 2003: A Cyclone Phase Space Derived from Thermal Wind and Thermal Asymmetry. *Mon. Wea. Rev.*, **131**, 585–616,
[https://doi.org/10.1175/1520-0493\(2003\)131<0585:ACPSDF>2.0.CO;2](https://doi.org/10.1175/1520-0493(2003)131<0585:ACPSDF>2.0.CO;2).

Marchok, T., 2021: Important factors in the tracking of tropical cyclones in operational models. *J. Appl. Meteor. Climatol.*, <https://doi.org/10.1175/JAMC-D-20-0175.1>.